



Top 75 CAT Data change over a period Questions With Video Solutions

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Questions

Instructions

Answer the questions based on the following information. The amount of money invested (rupees in crores) in the core infrastructure areas of two districts, Chittoor and Khammam, Andhra Pradesh, is as follows.

	Chittoor district			Khammam District	
Core Area	1995	1996	Core area	1995	1996
Electricity	815.2	1054.2	Electricity	2065.8	2365.1
Chemical	389.5	476.7	Chemical	745.3	986.4
Thermal	632.4	565.9	Thermal	1232.7	1026.3
Solar	468.1	589.6	Solar	1363.5	1792.1
Nuclear	617.9	803.1	Nuclear	1674.3	2182.1
Total	2923.1	3489.5	Total	7081.6	8352

Question 1

The investment in electricity and thermal energy in 1995 in these two districts formed what per cent of the total investment made in that year?

- A 41%
- B 47%
- C 52%
- D 55%

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Question 2

If the total investment in Khammam shows the same rate of increase in 1997, as it had shown from 1995 to 1996, what approximately would be the total investment in Khammam in 1997?

- A Rs. 9,850 crore
- B Rs. 10,020 crore
- C Rs. 9,170 crore
- D Rs. 8,540 crore

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Question 3

In Khammam district, the investment in which area in 1996 showed the highest percentage increase over the investment in that area in 1995?

- A Electricity
- B Chemical
- C Solar
- D Nuclear



Question 4

By what per cent was the total investment in the two districts more in 1996 as compared to 1995?

- A 14%
- B 21%
- C 24%
- D 18%

Question 5

Approximately how many times was the total investment in Chittoor to the total investment in Khammam?

- A 2.8
- B 2
- C 2.4
- D 1.7

Instructions

DIRECTIONS for the following three questions: Answer the questions on the basis of the information given below.

Date of Issue	Notified Amount	Maturity	Competitive bids received	Non-competitive bids received	Competitive bids accepted		Non competitive bids accepted		Total Amount	Coupon Rate %	Implicit Yield %
	Rs. Crore	Years	No.	No.	No.	Value	No.	Value	Rs. Crore		
17-Jul-02	40	15	229	23	66	15.21	23	0.37	16	8.07	7.80
17-Jul-02	30	10	145	12	90	29.88	12	0.12	30	6.72	6.72
5-Aug-02	50	9	324	13	105	49.68	13	0.33	50	9.39	7.24
5-Aug-02	20	24	163	9	34	19.81	9	0.19	20	10.18	7.93
28-Aug-02	50	15	260	26	157	48.92	26	1.08	50	7.46	7.46
28-Aug-02	20	30	119	15	67	19.61	15	0.39	20	7.95	7.95
11-Sep-02	40	15	261	22	152	38.93	22	1.07	40	7.46	7.44
11-Sep-02	30	20	131	20	98	29.44	20	0.56	30	8.35	7.70
9-Oct-02	40	11	361	26	119	39.22	26	0.78	40	7.27	7.14
9-Oct-02	30	30	91	15	39	29.52	15	0.48	30	7.95	7.89
7-Nov-02	40	17	245	14	20	39.71	14	0.29	40	10.03	7.26
7-Nov-02	30	24	166	11	49	29.70	11	0.31	30	10.18	7.48
9-Apr-03	40	20	245	25	65	39.53	25	1.47	40	6.30	6.30
9-Apr-03	50	11	236	24	201	49.40	24	0.60	50	7.37	5.98
23-Apr-03	50	15	319	26	134	48.98	26	1.02	50	6.25	6.10
23-Apr-03	20	29	131	19	9	19.39	19	0.61	20	7.95	6.33
5-May-03	60	10	314	14	98	59.69	14	0.31	60	7.27	5.97
5-May-03	30	20	143	14	118	29.58	14	0.42	30	6.30	6.35
4-Jun-03	30	25	187	19	15	28.50	19	1.50	30	6.13	6.13
4-Jun-03	60	9	378	21	151	59.09	21	0.91	60	6.85	5.76
2-Jul-03	50	11	298	20	116	49.05	20	0.95	50	7.37	5.76
2-Jul-03	30	25	114	20	45	28.64	20	1.36	30	6.31	6.10
16-Jul-03	60	17	371	29	115	57.00	29	3.10	60	6.35	5.97
16-Jul-03	30	29	134	22	12	29.32	22	0.68	30	7.95	6.20
Total	930								906		

One of the functions of the Reserve Bank of India is to mobilize funds for the Government of India by issuing securities. The following table shows details of funds mobilized during the period July 2002 - July 2003. Notice that on each date there were two rounds of issues, each with a different maturity.

Question 6

Which of the following statements is NOT true?

- A Competitive bids received always exceed non-competitive bids received.
- B The number of competitive bids accepted does not always exceed the number of non-competitive bids accepted.
- C The value of competitive bids accepted on any particular date is never higher for higher maturity.
- D The value of non-competitive bids accepted in the first round is always greater than that in the second round.

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Question 7

Which of the following is true?

- A The second round issues have a higher maturity than the first round for all dates.
- B The second round issue of any date has a lower maturity only when the first round notified amount exceeds that of the second round.
- C On at least one occasion, the second round issue having lower maturity received a higher number of competitive bids.
- D None of the above three statements is true.

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Question 8

How many times was the issue of securities under-subscribed, i.e., how often did the total amount mobilized fall short of the amount notified?

- A 0
- B 1
- C 2
- D 3

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Instructions

DIRECTIONS for the following three questions: Answer the questions on the basis of the information given below.

Spam that enters our electronic mailboxes can be classified under several spam heads. The following table shows the distribution of such spam worldwide over time. The total number of spam emails received during December 2002 was larger than the number received in June 2003. The total number of spam emails received during September 2002 was larger than the number received in March 2003. The figures in the table represent the percentage of all spam emails received during that period, falling into those respective categories.

Category	Sep-02	Dec-02	Mar-03	Jun-03
Adult	38	33	19	17
Financial	25	30	37	45
Health	11	19	5	18
Internet	5	3	10	6
Products	3	7	10	11
Scams	5	6	11	2
Others	13	2	8	1

Question 9

In the health category, the number of spam emails received in December 2002 as compared to June 2003.

- A was larger
- B was smaller
- C was equal
- D cannot be determined

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Question 10

In the financial category, the number of spam emails received in September 2002 as compared to March 2003.

- A was larger
- B was smaller
- C was equal
- D cannot be determined

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Question 11

In which category was the percentage of spam emails increasing but at a decreasing rate?

- A Financial
- B Scams
- C Products
- D None of the above

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Instructions

Directions for the next 4 questions: Answer these questions based on the table below:

The table shows trends in external transactions of Indian corporate sector during the period 1993-94 to 1997-98. In addition, following definitions hold good:

Sales, Imports, and Exports, respectively denote the sales, imports and exports in year i .

Deficit in year i , Deficit $_i$ = Imports - Exports

Deficit Intensity in year i , DI = Deficit/Sales Growth rate of deficit intensity in year i , $GDI = \frac{DI_i - DI_{i-1}}{DI_{i-1}}$

Further, note that all imports are classified as either raw material or capital goods.

Trends in External Transactions of Indian Corporate Sector (All figures in %)

Year	1997-98	1996-97	1995-96	1994-95	1993-94
Export Intensity*	9.2	8.2	7.9	7.5	7.3
Import Intensity*	14.2	16.2	15.5	13.8	12.4
Imported raw material/total cost of raw material	20.2	19.2	17.6	16.3	16
Imported capital goods/Gross fixed assets	17.6	9.8	11.8	16.3	19.5

*Ratio of exports (or imports) to sales

Question 12

Which of the following statements can be inferred to be true from the given data?

- A During the 5 year period between 1993-94 and 1997-98, exports have increased every year.
- B During the 5 year period between 1993-94 and 1997-98, imports have decreased every year.
- C Deficit in 1997-98 was lower than that in 1993-94.
- D Deficit intensity has increased every year between 1993-94 and 1996-97.

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Question 13

In 1997-98 the total cost of raw materials is estimated as 50% of sales of that year. The turn over of Gross fixed assets, defined as the ratio of sales to Gross fixed assets, in 1997-98 is, approximately;

- A 3.3
- B 4.3
- C 0.33
- D not possible to determine

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Question 14

The highest growth rate in deficit intensity was recorded in:

- A 1994-95
- B 1995-96
- C 1996-97
- D 1997-98

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Question 15

The value of the highest growth rate in deficit intensity is approximately:

- A 8.45%
- B 2.15%
- C 33.3%
- D 23.5%

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Instructions

Directions for the next 5 questions: Answer the questions with reference to the table given below: Information Technology Industry in India (Figures are in million US dollars)

	1994-95	1995-96	1996-97	1997-98	1998-99
Software:					
Domestic	350	490	670	950	1250
Exports	485	734	1083	1750	2650
Hardware:					
Domestic	590	1037	1050	1205	1026
Exports	177	35	286	201	4
Peripherals:					
Domestic	148	196	181	229	329
Exports	6	6	14	19	18
Training	107	143	185	263	302
Maintenance	142	172	182	221	236
Networking and others	36	73	156	193	237
Total	2041	2886	3807	5031	6052

Question 16

The highest percentage growth in the total IT business, relative to the previous year was achieved in:

- A 1995-96
- B 1996-97
- C 1997-98
- D 1998-99

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Question 17

Additional instructions:

For any activity A, year X dominates years Y if IT business in activity A, in the year X, is greater than the IT business, in activity A, in the year Y. For any two IT business activities, A & B, year X dominates year Y if:

- a) The IT business in activity A, in the year X, is greater than or equal to the IT business, in activity A in the year Y,
- b) The IT business in activity B, in the year X, is greater than or equal to the IT business in activity B in the year Y and
- c) There should be strict inequality in the case of at least one activity.

For the IT hardware business activity, which one of the following is not true?

- A 1997-98 dominates 1996-97
- B 1997-98 dominates 1995-96
- C 1995-96 dominates 1998-99
- D 1998-99 dominates 1996-97

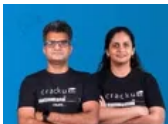
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Question 18

The total annual exports lay between 35 and 40 percent of the total annual business of the IT industry, in the years:

- A 1997-98 & 1994-95
- B 1996-97 & 1997-98
- C 1996-97 & 1998-99
- D 1996-97 & 1994-95

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IIM Call Predictor

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Question 19

Which one of the following statements is correct?

- A The annual software exports steadily increased but annual hardware exports steadily declined during 1994-1999.
- B The annual peripheral exports steadily increased during 1994-1999.
- C The IT business in training during 1994- 1999 was higher than the total IT business in maintenance during the same period.
- D None of the above statements is true.

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Question 20

Additional instructions:

For any activity A, year X dominates years Y if IT business in activity A, in the year X, is greater than the IT business, in activity A, in the year Y. For any two IT business activities, A & B, year X dominates year Y if:

- a) The IT business in activity A, in the year X, is greater than or equal to the IT business, in activity A in the year Y,
- b) The IT business in activity B, in the year X, is greater than or equal to the IT business in activity B in the year Y and
- c) There should be strict inequality in the case of at least one activity.

For the two IT business activities, hardware and peripherals, which one of the following is true?

- A 1996-97 dominates 1995-96
- B 1998-99 dominates 1995-96
- C 1997-98 dominates 1998-99
- D None of these

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Instructions

The following table gives the national income and the population of a country for the years 1984 – 85 to 1989 – 90. For each of the following questions choose the best alternative:

Year	National Income (in Rs. crore)	Population (in crore)
1984-85	229,225	74
1985-86	261,174	75
1986-87	291,556	77
1987-88	329,934	78.5
1988-89	388,539	80
1989-90	433,500	81.2

Question 21

The per capita income is highest for the year :

- A 1984-85
- B 1985-86
- C 1987-88
- D 1989-90

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CAT Percentile Predictor



Question 22

Increase in the per capita income compared to the previous year among the years given below was highest for the year:

- A 1985-86
- B 1986-87
- C 1987-88
- D 1988-89

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Question 23

The difference between the percentage increase in per capita income and the percentage increase in the population compared to the previous year is highest for the year:

- A 1985-86
- B 1986-87
- C 1987-88
- D 1988-89

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Question 24

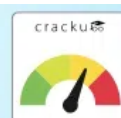
The rate of increase in population was lowest in the year:

- A 1985-86
- B 1987-88
- C 1989-90
- D None of these

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CAT Score Calculator



Question 25

The increase in the per capita income compared to the previous year is lowest for the year :

- A 1985-86
- B 1986-87
- C 1987-88
- D 1989-90

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Instructions

The following table gives the number of rooms in different hotels in Mumbai and the cost of building each one of them. Note that all the projects started in 1997

Project	No. of rooms	Cost (Rs. In crore)	Year of Completion	Company
Windsor Manor	600	275	1999	IHCL
Leela Hotels	310	235	1999	Leela Hotels
Mumbai Heights	250	250	1998	Bombay Hotels
Royal Holidays	536	225	1998	Lokhandwala Groups
Majestic Holiday	500	250	1999	Raheja Group
Supremo Hotel	300	300	1999	ITC
Hyatt Regency	500	250	2000	Asian Hotels

Note: All projects start in 1997.

Question 26

Which of the following has the maximum number of rooms per crore of rupees?

- A IHCL
- B Raheja Group
- C Lokhandwala Group
- D ITC

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Question 27

Which of the following had the least cost per room?

- A Lokhandwala Group
- B Raheja Group
- C IHCL
- D ITC

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Question 28

Assume that the cost of the project is incurred in the year of completion; interest is charged at the rate of 10% per annum. What is the cost incurred for projects completed in 1998?

- A Rs. 475 crore
- B Rs. 500 crore
- C Rs. 522.5 crore
- D Rs. 502.5 crore

[VIEW SOLUTION](#)

Question 29

What is the approximate cost incurred for projects completed by 2000? (in crores)

- A 1,785
- B 2,140
- C 2,320
- D None of these

[VIEW SOLUTION](#)

Question 30

Assume that the cost of the project is incurred in the year of completion; interest is charged at the rate of 10% per annum. What is the cost incurred for projects completed in 1999?

- A Rs. 1,282.6 crore
- B Rs. 1,270 crore
- C Rs. 1,805.1 crore
- D Rs. 1,535 crore

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Instructions

The following table gives the tariff [in paise per kilo-watt-hour (kWh)] levied by the UPSEB in 1994-95, in four sectors and the regions within them. The table also gives the percentage change in the tariff as compared to 1991-92.

	Region 1		Region 2		Region 3		Region 4		Region 5	
	P/kWh	%	P/kWh	% incr.	P/kWh	%incr.	P/kWh	% incr.	P/kWh	% incr.
Sector 1	425	15	472	5	420	-4	415	8	440	10
Sector 2	430	12	468	8	448	7	423	-3	427	11
Sector 3	428	8	478	-4	432	6	441	10	439	8
Sector 4	434	-5	470	15	456	10	451	12	446	-12

Question 31

In 1994-95, if there was 10% decrease in the domestic consumption of power as compared to that in 1991-92, what was the consumption of power in the rural sector in 1991-92?

- A 1,312 megawatts
- B 1,422 megawatts
- C 1,750 megawatts
- D Cannot be determined

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Question 32

If the amount of power consumed by the various regions in sector 1 is the same, then as compared to 1991-92 the net tariff in 1994-95

- A increased by 6.5%
- B decreased by 3.5%
- C increased by 10.2%
- D decreased by 7.3%

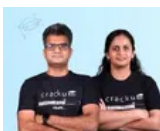
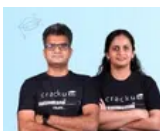
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Question 33

Which of the following statements is true?

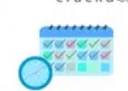
- A The average tariff in region 4 is 437.5 p/kWh
- B The average tariff in region 2 is greater than the average tariff in region 5
- C In 1991-92, the industrial sector contributed to about 42% of the total revenue from power
- D None of these

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Question 34

In the given 2 years, what is the total tariff paid by the urban sector?

- A Rs. 22.4 lakh
- B Rs. 21.6 lakh
- C Rs. 27.2 lakh
- D Cannot be determined

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Question 35

What was the approximate average tariff in sector 3 in 1991-92?

- A 407
- B 420
- C 429
- D None of these

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Instructions

The table given below gives the annual details of loans from rural banks and agricultural loans over the years 1970 to 1983. Using this data answer the questions that follow.

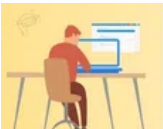
Year	Loan from Rural			Agricultural Loans		
	No. of rural banks	Avg. no. of loans	Average size (in Rs.)	No. ('000)	Value (Rs. In millions)	Consumer Price Index
1970	90	28	109	18.3	2	43
1971	115	39	133	20.4	3.58	49
1972	130	52	178	25.1	6.26	55
1974	260	98	243	41.2	34.54	70
1975	318	121	283	51.4	52.21	78
1980	605	288	567	135.7	498.4	131
1981	665	312	622	152.8	612.4	137
1983	840	380	711	211.6	915.7	149

Question 36

If the consumer price index for 1970 is to be taken as 105 and the indices for the subsequent years are to be corrected accordingly, what is the approximate value of the agricultural loans in 1980 at 1983 prices?

- A Rs. 570 million
- B Rs. 680 million
- C Rs. 525 million
- D Rs. 440 million

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Question 37

Which of the following pairs of years showed the maximum increase in the number of rural bank loans?

- A 1971-72
- B 1974-75
- C 1970-71

D 1980-81

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Question 38

From the given data, the number of rural loans up to 1980 formed approximately what percentage of those in 1983?

- A 112%
- B 80%
- C 97%
- D Cannot be determined

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Question 39

What is the simple annual rate of increase in the number of agricultural loans from 1970 to 1983?

- A 132%
- B 81%
- C 75%
- D 1056%

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Question 40

In which year was the number of rural bank loans per rural bank least?

- A 1974
- B 1971
- C 1970
- D 1975

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Question 41

If the consumer price index for 1970 is to be taken as 105 and the indices for the subsequent years are to be corrected accordingly, by roughly how many points do the indices for 1983 and 1975 differ?

- A 174

- B 180
- C 188
- D 195

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Question 42

What is the value(in millions) of the agricultural loans in 1983 at 1970 prices?

- A Rs. 326
- B Rs. 264
- C Rs. 305
- D None of these

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Question 43

In 1974, the amount of agricultural loans formed what percentage of the total loans?

- A 85%
- B 71%
- C 77%
- D Cannot be determined

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Instructions

For the following questions answer them individually

Question 44

The table below shows the age-wise distribution of the population of Reposia. The number of people aged below 35 years is 400 million.

Age Group	Percentage
Below 15 years	30.00
15-24	17.75
25-34	17.00
35-44	14.50
45-54	12.50
55-64	7.10
65 and above	1.15

If the ratio of females to males in the 'below 15 years' age group is 0.96, then what is the number of females (in millions) in that age group?

- A 82.8
- B 90.8
- C 80.0
- D 90.0

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Instructions

Direction for questions: Answer the questions based on the following information. The data given in the table shows the investment details in country 'Fortune Land' of companies A, B, C, D, E and F. Figures in the table are in US dollars in billions.

	A	B	C	D	E	F
year 1	2.5	4.6	5.8	3.11	10.6	7.8
year 2	6.7	7.5	12.5	5.6	17.4	25.3
Year 3	11.5	18.7	21.2	7.7	29.8	60.1

Question 45

For which company is investment not increased from year 1 to year 3?

- A C
- B D
- C F
- D None of these

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Question 46

By what percentage is the investments of companies D,E,F greater than the investments of companies A, B, C in year 2?

- A 75%
- B 81%
- C 67.5%
- D 42.3%

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Question 47

What is the ratio of investments of E to F for the years 1 to 3?

- A 31 : 19

B 19 : 31

C 20 : 29

D 41 : 53

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Question 48

What is the percentage increase in investment of B, C, D and E from year 1 to year 3?

A 121%

B 321%

C 221%

D 300%

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100+ CAT DILR Questions



Question 49

What is D's contribution as a percentage of total investments in year 2?

A 8.2%

B 4.5%

C 7.4%

D 9.2%

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Instructions

The following table shows the revenue and expenses in millions of Euros (European currency) associated with REPSOL YPF company's oil and gas producing activities in operations in different parts of the world for 1998-2000.

S. No.	Item	Year	Total World	Spain	North Africa & Middle East	Argentina	Rest of Latin America	Far East	North Sea	Rest of the World
1	Revenue	1998	916	70	366	281	34	82	78	5
		1999	3374	55	666	2006	115	301	140	91
		2000	8328	394	1290	5539	482	603	0	20
2	Expenses	1998	668	39	255	187	57	63	52	15
		1999	1999	48	325	1168	131	204	65	58
		2000	3709	43	530	2540	252	311	0	33
3	Income Before Taxes & Charges (Revenue - Expenses)	1998	248	31	111	94	-23	19	26	-10
		1999	1375	7	341	838	-16	97	75	33
		2000	4619	351	760	2999	230	292	0	-13
4	Taxes and Charges	1998	152	6	104	33	-3	9	6	-3
		1999	561	3	169	338	-6	39	21	-3
		2000	1845	126	404	1150	61	103	0	1
5	Net Income = (3)-(4)	1998	96	25	7	61	-20	10	20	-7
		1999	814	4	172	500	-10	58	54	36
		2000	2774	225	356	1849	169	189	0	-14

Question 50

If efficiency is defined as the ratio of revenue to expenses, which operation was the least efficient in 2000?

- A Spain
- B Argentina
- C Far East
- D None of these

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Question 51

In 2000, which among the following countries had the best profitability?

- A North Africa and Middle East
- B Spain
- C Rest of Latin America
- D Far East

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100+ CAT VARC Questions



Question 52

How many operations registered a sustained yearly increase in income before taxes and charges from 1998 to 2000?

- A 3
- B 4
- C 5
- D None of these

[VIEW SOLUTION](#)

Question 53

How many operations (Spain, North Africa and Middle East...) of the company witnessed more than 200% increase in revenue from 1999 to 2000?

- A 1
- B 2
- C 3
- D None of these

[VIEW SOLUTION](#)

Question 54

Of the following statements, which one is not true?

- A The operations in Spain had the best efficiency in 2000
- B The Far East operations witnessed an efficiency improvement from 1999 to 2000
- C The North Sea operations witnessed an efficiency improvement from 1998 to 1999
- D In 1998, the operations in Rest of Latin America were the least efficient

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CAT Expected Questions PDF



Question 55

How many operations (Spain, North Africa and Middle East,..) of the company accounted for less than 5% of the total revenue earned in 1999?

- A 2
- B 3
- C 4
- D None of these

[VIEW SOLUTION](#)

Question 56

If profitability is defined as the ratio of net income after taxes and charges to expense, which of the following statements is true?

- A The Far East operations witnessed its highest profitability in 1998
- B The North Sea operations increased its profitability from 1998 to 1999
- C The operations in Argentina witnessed a decrease in profitability from 1998 to 1999
- D Both 2 and 3 are true

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Question 57

Ignoring the loss making operations of the company in 1998, for how many operations was the percentage increase in net income before taxes and charges higher than the average from 1998 to 1999?

- A 0
- B 1

C 2

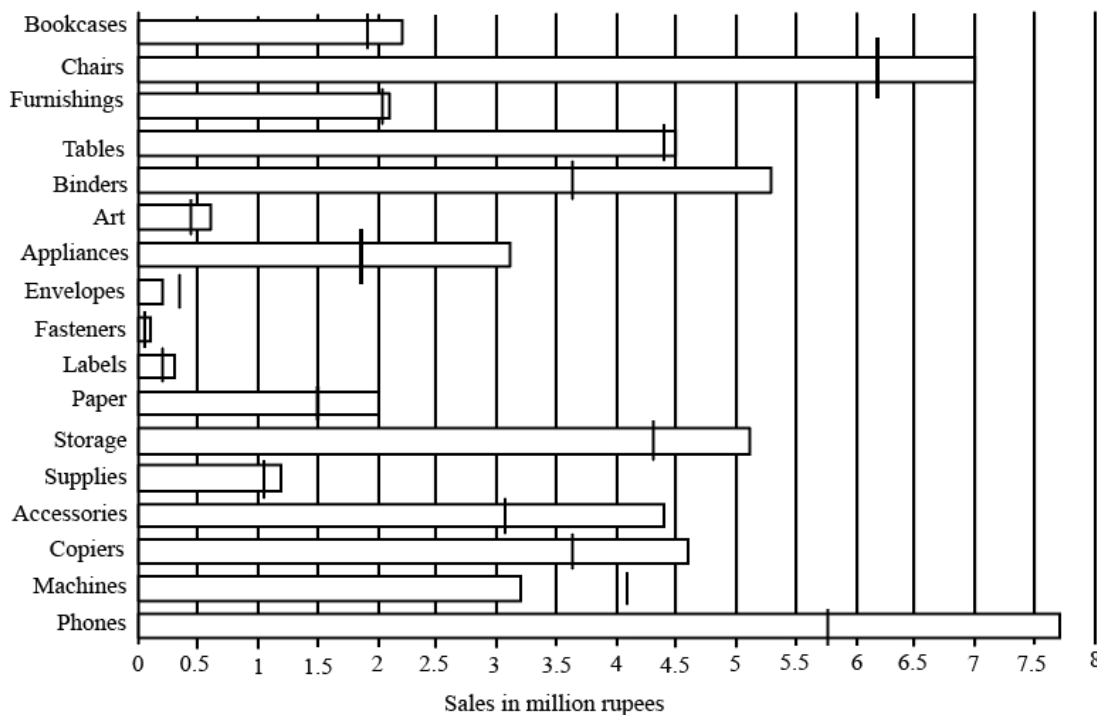
D None of these

[VIEW SOLUTION](#)

CAT Important Topics

Instructions

Sub-category



The horizontal bars in the above diagram represent 2020 aggregate sales (in ₹ million) of a company for the different subcategories of its products. The top four product subcategories (Bookcases, Chairs, Furnishings, Tables) belong to furniture product category; the bottom four product subcategories (Accessories, Copiers, Machines, Phones) belong to the technology product category while all other product subcategories belong to the office supply product category. For each of the product subcategories, there is a vertical line indicating the sales of the corresponding subcategory in 2019.

Question 58

The improvement index for a category is the maximum percentage increase in sales from 2019 to 2020 among any of its subcategories. The correct order of categories in increasing order of this improvement index is

- A furniture, technology, office supply
- B technology, furniture, office supply
- C office supply, technology, furniture
- D office supply, furniture, technology



 VIDEO SOLUTION

Question 59

The total sales (in ₹ million) in 2019 from products in office supplies category is closest to

- A 18.0
- B 16.5
- C 13.5
- D 12.5



 VIDEO SOLUTION

Question 60

How many subcategories had sales of ₹ 4 million or more in 2019 and registered an increase in sales in excess of 25% in 2020?



VIDEO SOLUTION



Question 61

The percentage increase in sales in Furniture category from 2019 to 2020 is closest to

- A 20%
- B 8%
- C 25%
- D 1%



VIDEO SOLUTION

Instructions

There are 21 employees working in a division, out of whom 10 are special-skilled employees (SE) and the remaining are regular-skilled employees (RE). During the next five months, the division has to complete five projects every month. Out of the 25 projects, 5 projects are "challenging", while the remaining ones are "standard". Each of the challenging projects has to be completed in different months. Every month, five teams – T1, T2, T3, T4 and T5, work on one project each. T1, T2, T3, T4 and T5 are allotted the challenging project in the first, second, third, fourth and fifth month, respectively. The team assigned the challenging project has one more employee than the rest.

In the first month, T1 has one more SE than T2, T2 has one more SE than T3, T3 has one more SE than T4, and

T4 has one more SE than T5. Between two successive months, the composition of the teams changes as follows:

- The team allotted the challenging project, gets two SE from the team which was allotted the challenging project in the previous month. In exchange, one RE is shifted from the former team to the latter team.
- After the above exchange, if T1 has any SE and T5 has any RE, then one SE is shifted from T1 to T5, and one RE is shifted from T5 to T1. Also, if T2 has any SE and T4 has any RE, then one SE is shifted from T2 to T4, and one RE is shifted from T4 to T2.

Each standard project has a total of 100 credit points, while each challenging project has 200 credit points. The credit points are equally shared between the employees included in that team.

Question 62

Which of the following CANNOT be the total credit points earned by any employee from the projects?

- A 140
- B 150
- C 170
- D 200



There are 21 employees working in a division, out of whom 10 are special-skilled employees (SE) and the remaining are regular-skilled employees (RE). During the next five months, the division has to complete five projects every month. Out of the 25 projects, 5 projects are "challenging", while the remaining ones are "standard". Each of the challenging projects has to be completed in different months. Every month, five teams – T1, T2, T3, T4 and T5 – work on one project each. T1, T2, T3, T4 and T5 are allotted the challenging project in the first, second, third, fourth and fifth month, respectively. The team assigned the challenging project has one more employee than the rest of the teams. In the first month, T1 has one more SE than T2, T2 has one more SE than T3, T3 has one more SE than T4, and T4 has one more SE than T5.

$$x + x + 1 + x + 2 + x + 3 + x + 4 = 10$$
$$5x + 10 = 10$$
$$x = 0$$

VIDEO SOLUTION

Question 63

The number of times in which the composition of team T2 and the number of times in which composition of team T4 remained unchanged in two successive months are:

- A (2,1)
- B (1,0)
- C (0,0)
- D (1,1)

cracku

There are 21 employees working in a division, out of whom 10 are special-skilled employees (SE) and the remaining are regular-skilled employees (RE). During the next five months, the division has to complete five projects every month. Out of the 25 projects, 5 projects are "challenging", while the remaining ones are "standard". Each of the challenging projects has to be completed in different months. Every month, five teams – T1, T2, T3, T4 and T5, work on one project each. T1, T2, T3, T4 and T5 are allotted the challenging project in the first, second, third, fourth and fifth month, respectively. The team assigned to a challenging project has one more employee than the rest of the teams. In the first month, T1 has one more SE than T2, T2 has one more SE than T3, T3 has one more SE than T4, and T4 has one more SE than T5.

$$x + x + 1 + x + 2 + x + 3 + x + 4 = 10$$

$$5x + 10 = 10$$

$$x = 0$$

VIDEO SOLUTION

cracku

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Question 64

The number of SE in T1 and T5 for the projects in the third month are, respectively:

- A (0,2)
- B (0,3)
- C (1,2)
- D (1,3)

cracku

There are 21 employees working in a division, out of whom 10 are special-skilled employees (SE) and the remaining are regular-skilled employees (RE). During the next five months, the division has to complete five projects every month. Out of the 25 projects, 5 projects are "challenging", while the remaining ones are "standard". Each of the challenging projects has to be completed in different months. Every month, five teams – T1, T2, T3, T4 and T5, work on one project each. T1, T2, T3, T4 and T5 are allotted the challenging project in the first, second, third, fourth and fifth month, respectively. The team assigned to a challenging project has one more employee than the rest of the teams. In the first month, T1 has one more SE than T2, T2 has one more SE than T3, T3 has one more SE than T4, and T4 has one more SE than T5.

$$x + x + 1 + x + 2 + x + 3 + x + 4 = 10$$

$$5x + 10 = 10$$

$$x = 0$$

VIDEO SOLUTION

Question 65

One of the employees named Aneek scored 185 points. Which of the following CANNOT be true?

- A Aneek worked only in teams T1, T2, T3, and T4.
- B Aneek worked only in teams T1, T2, T4, and T5.
- C Aneek worked only in teams T2, T3, T4, and T5.
- D Aneek worked only in teams T1, T3, T4, and T5.

cracku

There are 21 employees working in a division, out of whom 10 are special-skilled employees (SE) and the remaining are regular-skilled employees (RE). During the next five months, the division has to complete five projects every month. Out of the 25 projects, 5 projects are "challenging", while the remaining ones are "standard". Each of the challenging projects has to be completed in different months. Every month, five teams – T1, T2, T3, T4 and T5, work on one project each. T1, T2, T3, T4 and T5 are allotted the challenging project in the first, second, third, fourth and fifth month, respectively. The team assigned to a challenging project has one more employee than the rest of the teams. In the first month, T1 has one more SE than T2, T2 has one more SE than T3, T3 has one more SE than T4, and T4 has one more SE than T5.

$$x + x + 1 + x + 2 + x + 3 + x + 4 = 10$$

$$5x + 10 = 10$$

$$x = 0$$

VIDEO SOLUTION

Instructions

The table below shows the estimated cost (in Rs. Lakh) of a project of laying a railway line between two places.

	1988	1989	1990	1991
Surveying	41.5	7.5	2.2	0.5
Cement	----	95	80	75
Steel	----	70	45	60
Bricks	----	15	12	16
Building materials	----	25	18	21
Labour	2.1	25	20	18
Administration	7.5	15	15	14
Contingencies	1	15	4.2	5
Total	52.1	267.5	196.4	209.5

Question 66

If the cost of materials ((cement, steel, bricks, building materials) rises by 5% each year from 1990 onwards, by how much will the estimated cost rise for the years 1990 and 1991?

- A Rs. 11.45 lakh
- B Rs. 16.35 lakh
- C Rs. 21.45 lakh
- D Rs. 26.45 lakh

The table below shows the estimated cost (in Rs. Lakh) of a project of laying a railway line between two places.

	1988	1989	1990	1991
Surveying	41.5	7.5	2.2	0.5
Cement	—	95	80	75
Steel	—	70	45	60
Bricks	—	15	12	18
Building materials	—	25	18	23
Labour	2.1	25	20	18
Administration	7.5	15	15	15
Contingencies	1	15	4.2	—
Total	52.1	267.5	196.4	99.5

The total expenditure is required to be kept within Rs. 326 lakh by cutting down on administration equally in all the years. What will be the percentage cut for 1989?

A) 22.6
B) 32.6
C) 42.5
D) 52.6

Handwritten calculation: $\frac{25.5}{4}$

VIDEO SOLUTION

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Question 67

If the length of the line to be laid each year is proportional to the estimated cost for cement, steel, bricks, building material and labour, what fraction of the total length is proposed to be completed by the third year?

- A 0.9
- B 0.7
- C 0.6
- D 0.3

The table below shows the estimated cost (in Rs. Lakh) of a project of laying a railway line between two places.

	1988	1989	1990	1991
Surveying	41.5	7.5	2.2	0.5
Cement	—	95	80	75
Steel	—	70	45	60
Bricks	—	15	12	18
Building materials	—	25	18	23
Labour	2.1	25	20	18
Administration	7.5	15	15	15
Contingencies	1	15	4.2	—
Total	52.1	267.5	196.4	99.5

The total expenditure is required to be kept within Rs. 326 lakh by cutting down on administration equally in all the years. What will be the percentage cut for 1989?

A) 22.6
B) 32.6
C) 42.5
D) 52.6

Handwritten calculation: $\frac{25.5}{4}$

VIDEO SOLUTION

Question 68

After preparing the estimate, the provision for contingencies is felt inadequate and is therefore doubled. By what percent does the total estimate increase?

- A 3.47
- B 2.45

C 1.50

D 3.62

The table below shows the estimated cost (in Rs. Lakh) of a project of laying a railway line between two places.

	1988	1989	1990	1991
Surveying	41.5	7.5	2.2	0.5
Cement	95	80	75	
Steel	70	45	60	
Bricks	15	12	18	
Building materials	25	18	23	
Labour	2.1	25	20	18
Administration	7.5	15	15	15
Contingencies	1	15	4.2	
Total	52.1	267.5	196.4	39.5

The total expenditure is required to be kept within Rs. 32.6 lakh to cut on administration equally in all the years. What will be the percentage cut for 1989?

A) 22.6
B) 32.6
C) 42.5
D) 52.6

Handwritten calculation: $\frac{25.5}{4}$

VIDEO SOLUTION

Question 69

It is found at the end of 1990, that the entire amount estimated for the project has been spent. If for 1991, the actual amount spent was equal to that which was estimated, by what percent (approximately) has the actual expenditure exceeded the estimated expenditure?

A 39

B 29

C 19

D 9

The table below shows the estimated cost (in Rs. Lakh) of a project of laying a railway line between two places.

	1988	1989	1990	1991
Surveying	41.5	7.5	2.2	0.5
Cement	95	80	75	
Steel	70	45	60	
Bricks	15	12	18	
Building materials	25	18	23	
Labour	2.1	25	20	18
Administration	7.5	15	15	15
Contingencies	1	15	4.2	
Total	52.1	267.5	196.4	39.5

The total expenditure is required to be kept within Rs. 32.6 lakh to cut on administration equally in all the years. What will be the percentage cut for 1989?

A) 22.6
B) 32.6
C) 42.5
D) 52.6

Handwritten calculation: $\frac{25.5}{4}$

VIDEO SOLUTION

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Question 70

What is the approximate ratio of the total cost of materials (cement, steel, bricks, building materials) for all the years to the total labour cost?

- A 4 : 1
- B 8 : 1
- C 12 : 1
- D 16 : 1

The table below shows the estimated cost (in Rs. Lakh) of a project of laying a railway line between two places.

	1988	1989	1990	1991
Surveying	41.5	7.5	2.2	0.5
Cement	—	95	80	75
Steel	—	70	45	60
Bricks	—	15	12	10
Building materials	—	25	18	21
Labour	2.1	25	20	18
Administration	7.5	15	15	14
Contingencies	1	15	4.2	—
Total	52.1	267.5	186.4	99.5

The total expenditure is required to be kept within Rs. 700 lakh by cutting the expenditure on administration equally in all the years. What will be the percentage cut for 1989?

A) 22.6
B) 32.6
C) 42.5
D) 52.6

VIDEO SOLUTION

Question 71

The total expenditure is required to be kept within Rs. 700 lakh by cutting the expenditure on administration equally in all the years. What will be the percentage cut for 1989?

- A 22.6
- B 32.6
- C 42.5
- D 52.6

The table below shows the estimated cost (in Rs. Lakh) of a project of laying a railway line between two places.

	1988	1989	1990	1991
Surveying	41.5	7.5	2.2	0.5
Cement	—	95	80	75
Steel	—	70	45	60
Bricks	—	15	12	10
Building materials	—	25	18	21
Labour	2.1	25	20	18
Administration	7.5	15	15	14
Contingencies	1	15	4.2	—
Total	52.1	267.5	186.4	99.5

The total expenditure is required to be kept within Rs. 700 lakh by cutting the expenditure on administration equally in all the years. What will be the percentage cut for 1989?

A) 22.6
B) 32.6
C) 42.5
D) 52.6

VIDEO SOLUTION

Instructions

Directions for the following four questions:

Answer the following questions based on the information given below.

The following table shows the break-up of actual costs incurred by a company in last five years (year 2002 to year 2006) to produce a particular product.

	Year 2002	Year 2003	Year 2004	Year 2005	Year 2006
Volume of production and sale (Units)	1,000	900	1,100	1,200	1,200
Costs (Rs.)					
Material	50,000	45,100	55,200	59,900	60,000
Labour	20,000	18,000	22,100	24,150	24,000
Consumables	2,000	2,200	1,800	1,600	1,400
Rent of building	1,000	1,000	1,100	1,100	1,200
Rates and taxes	400	400	400	400	400
Repair and maintenance expenses	800	820	780	790	800
Operating cost of machines	30,000	27,000	33,500	36,020	36,000
Selling and marketing expenses	5,750	5,800	5,800	5,750	5,800

The production capacity of the company is 2000 units. The selling price for the year 2006 was Rs. 125 per unit. Some costs change almost in direct proportion to the change in volume of production, while others do not follow any obvious pattern of change with respect to the volume of production and hence are considered fixed. Using the information provided for the year 2006 as the basis for projecting the figures for the year 2007, answer the following questions:

Question 72

What is the approximate cost per unit in rupees, if the company produces and sells 1400 units in the year 2007?

- A 104
- B 107
- C 110
- D 115
- E 116

Directions for the following four questions:
Answer the following questions based on the information given below.

The following table shows the break-up of actual costs incurred by a company in last five years (year 2002 to year 2006) to produce a particular product.

	Year 2002	Year 2003	Year 2004	Year 2005	Year 2006
Volume of production and sale (Units)	1,000	900	1,100	1,200	1,200
Costs (Rs.)					
Material	50,000	45,100	55,200	59,900	60,000
Labour	20,000	18,000	22,100	24,150	24,000
Consumables	2,000	2,200	1,800	1,600	1,400
Rent of building	1,000	1,000	1,100	1,100	1,200
Rates and taxes	400	400	400	400	400
Repair and maintenance expenses	800	820	780	790	800
Operating cost of machines	30,000	27,000	33,500	36,020	36,000
Selling and marketing expenses	5,750	5,800	5,800	5,750	5,800

The production capacity of the company is 2000 units. The selling price for the year 2006 was Rs. 125 per unit. Some costs change almost in direct proportion to the change in volume of production, while others do not follow any obvious pattern of change with respect to the volume of production and hence are considered fixed. Using the information provided for the year 2006 as the basis for projecting the figures for the year 2007, answer the following questions:

What is the approximate cost per unit in rupees, if the company produces and sells 1400 units in the year 2007?

A) 104
B) 107
C) 110
D) 115
E) 116

VIDEO SOLUTION



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Question 73

If the company reduces the price by 5%, it can produce and sell as many units as it desires. How many units the company should produce to maximize its profit?

- A 1400

- B 1600
- C 1800
- D 1900
- E 2000

Directions for the following four questions:
Answer the following questions based on the information given below.

The following table shows the break-up of actual costs incurred by a company in last five years (year 2002 to year 2006) to produce a particular product.

	Year 2002	Year 2003	Year 2004	Year 2005	Year 2006
Volume of production and sale (units)	1,000	800	1,200	1,500	1,700
Costs (Rs.)					
Material	50,000	45,000	55,000	60,000	65,000
Labour	20,000	18,000	22,000	24,000	26,000
Consumables	2,000	2,000	1,800	1,800	1,800
Rent of building	1,000	1,000	1	1,100	1,200
Heats and lights	400	400	400	400	400
Repair and maintenance expenses	300	300	300	300	300
Operating cost of machines	10,000	11,000	12,000	13,000	14,000
Selling and marketing expenses	5,750	5,000	5,000	5,000	5,750

The production capacity of the company is 2000 units. The selling price is Rs. 125 per unit. Some costs change almost in direct proportion to the change in volume of production, while others do not. Using the information provided for the year 2006 as the base for proportion, answer the following questions:

What is the approximate cost per unit in rupees, if the company produces and sells 1400 units in the year 2007?

A) 104
B) 110
C) 116

VIDEO SOLUTION

Question 74

Given that the company cannot sell more than 1700 units, and it will have to reduce the price by Rs.5 for all units if it wants to sell more than 1400 units. What is the maximum profit that can be realized provided that the company sells more than 1400 units?

- A 25,400
- B 24,400
- C 31,400
- D 32,900
- E 32,000

Directions for the following four questions:
Answer the following questions based on the information given below.

The following table shows the break-up of actual costs incurred by a company in last five years (year 2002 to year 2006) to produce a particular product.

	Year 2002	Year 2003	Year 2004	Year 2005	Year 2006
Volume of production and sale (units)	1,000	800	1,200	1,500	1,700
Costs (Rs.)					
Material	50,000	45,000	55,000	60,000	65,000
Labour	20,000	18,000	22,000	24,000	26,000
Consumables	2,000	2,000	1,800	1,800	1,800
Rent of building	1,000	1,000	1	1,100	1,200
Heats and lights	400	400	400	400	400
Repair and maintenance expenses	300	300	300	300	300
Operating cost of machines	10,000	11,000	12,000	13,000	14,000
Selling and marketing expenses	5,750	5,000	5,000	5,000	5,750

The production capacity of the company is 2000 units. The selling price is Rs. 125 per unit. Some costs change almost in direct proportion to the change in volume of production, while others do not. Using the information provided for the year 2006 as the base for proportion, answer the following questions:

What is the approximate cost per unit in rupees, if the company produces and sells 1400 units in the year 2007?

A) 104
B) 110
C) 116

VIDEO SOLUTION

Question 75

What is the minimum number of units that the company needs to produce and sell to avoid any loss?

- A 313
- B 350
- C 384
- D 747
- E 928

Directions for the following four questions:
Answer the following questions based on the information given below.

The following table shows the break-up of actual costs incurred by a company in last five years (year 2002 to year 2006) to produce a particular product.

	Year 2002	Year 2003	Year 2004	Year 2005	Year 2006
Volume of production and sale (units)	1,000	900	1,500	1,200	1,400
Costs (Rs.)					
Material	50,000	45,000	80,000	70,000	60,000
Labour	20,000	18,000	22,000	14,000	24,000
Overhead	1,000	2,000	1,800	1,800	1,400
Cost of building	1,000	1,000	1,100	1,100	1,100
Power and steam	400	400	400	400	400
Repairs and maintenance expenses	800	800	800	800	800
Operating cost of machinery	10,000	12,000	10,000	10,000	10,000
Selling and marketing expenses	5,000	5,000	5,000	5,000	5,000

The production capacity of the company is 2000 units. The selling price of the product is Rs. 125 per unit. Some costs change almost in direct proportion to the change in volume of production, while others do not. Some costs change with respect to the volume of production and hence are considered fixed. Using the information provided for the year 2006 as the basis for proportionate figures for the year 2007, answer the following questions:

What is the approximate cost per unit in rupees, if the company produces and sells 1400 units in the year 2007?

A) 304
B) 310
C) 316
D) 322

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